

Derivatives and the Chain Rule

Mathematics Foundations — Session 3

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Today's Goal

- Read a derivative as an instantaneous rate of change, straight off a graph
 - Differentiate polynomials, exponentials, and logarithms
 - Apply the chain rule to a composed function, one layer at a time
 - See why the chain rule is backpropagation
- Deliverable: differentiate a composed function by hand and connect it to backprop in one sentence

Plan

1. Opening Quiz
2. The Derivative
3. Differentiation Rules
4. The Chain Rule
5. Backpropagation
6. Exercises
7. Recap
8. To Do for Next Session

Opening Quiz

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Opening Quiz: Matrix Multiplication & the Forward Pass

The Derivative

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- 2. The Derivative**
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Introduction

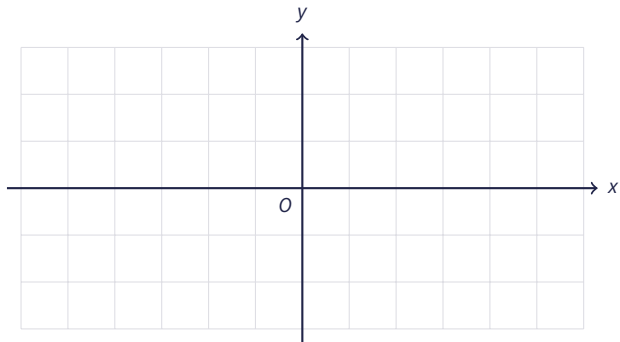
Recall: What Is a Function?

Vocabulary

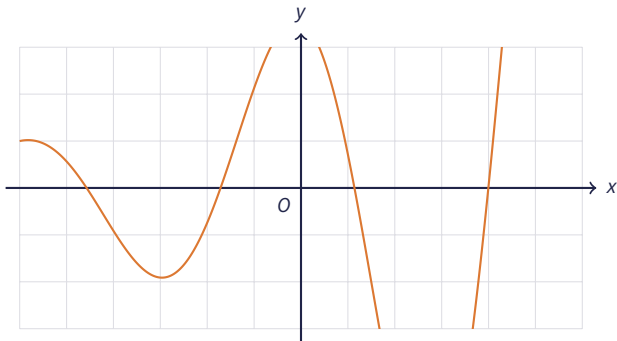
Function Composition

Function Composition: Example

Plotting a Curve



Rate of Change of a Curve



Definition of Speed

Speed: A Worked Example

Getting Back to Functions

One Remark

What Is the Effect of the Size of h ?

The Derivative

The Derivative: A Worked Example

Derivative Function

How Does a Computer Compute Derivatives?

Differentiation Rules

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Differentiation Rules

The Chain Rule

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The Chain Rule

Backpropagation

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Why the Chain Rule Is Backpropagation

Deliverable

Differentiate a composed function by hand, and write in one sentence how it connects to backpropagation.

Exercises

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Exercises

1. Differentiate $f(x) = 3x^2 + 5x - 2$
2. Differentiate $g(x) = e^{2x}$
3. Let $h(x) = (3x + 1)^2$. Using the chain rule, calculate $h'(x)$

Recap

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Recap

To Do for Next Session

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To Do for Next Session
